

LUMBINI ENGINEERING COLLEGE

Internal examination

Year: 2024

Full Marks:100

Pass Marks:45

Time :3hrs.

Level: Bachelor

Programme: BE (Computer)

Course: Calculus II

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

6x15=90

Attempt all the questions.

1a) Evaluate the given integral.

$$\int_0^{a/\sqrt{2}} \int_y^{\sqrt{a^2-y^2}} \log(x^2 + y^2) dx dy \text{ for } a>0, \text{ (by changing into polar form)} \quad \text{OR}$$

$$\iiint (x^2 z^2 + z^2 y^2 + y^2 x^2) dx dy dz \text{ throughout the volume of the sphere } x^2 + y^2 + z^2 = b^2.$$

b) Find the volume bounded by xy-plane, the paraboloid $2z = x^2 + y^2$ and the cylinder $x^2 + y^2 = 4$.

2a) Solve by using power series method. $(1+x^2)y''' + xy' - y = 0$.

b) What is Legendre's equation? Find it's general solution.

3a) Define Laplace transforms. State first and second shifting theorem of Laplace transform.

Find i) $L(t \cos wt)$ ii) $L^{-1}\left(\frac{se^{-25}}{s^2+1}\right)$

b) Solve the given initial value problem by using Laplace transform.

$$y'' + 2y' - 3y = 6e^{-2t}, y(0) = 2, y'(0) = -14$$

4a) Show that the value under the integral sign is exact in the plane and evaluate

$$\int_{(0,1,1/2)}^{(\pi/2,3,2)} [y^2 \cos x dx + (2y \sin x + e^{2z}) dy + 2ye^{2z} dz]$$

b) Evaluate $\iint \vec{F} \cdot \vec{n} dA$, Where $\vec{F} = (xy, yz, xy)$ and S is the part of the surface $x^2 + y^2 + z^2 = 1$, which lies in the first octant.

5a) Verify stoke's theorem for the vector function $\vec{F} = (y^3, 0, x^3)$ and C is the boundary of the triangle with vertices (1, 0, 0), (0, 1, 0) and (0, 0, 1).

b) Write the Fourier coefficient of a function $f(x)$. Find the Fourier expansion of the function $f(x) = x + |x|$ ($-\pi < x < \pi$).

6a) Find the Fourier sine as well as cosine series representation of the half range of the function $f(x) = x^2$ for $0 < x < L$.

b) Find the breaking time for $u_t + uu_x = 0$, $u(x, 0) = e^{-x^2}$.

4x2.5=10

7a) Find $f(t)$ for $\frac{2s+6}{s^2+4}$

b) Find the unit tangent vector at any point on the curve $x = 3\cos t, y = 3\sin t, z = 4t$.

c) Define even and odd function and find it for $x \cos nx$.

d) Define linear first order partial differential equation.



POKHARA UNIVERSITY
FACULTY OF SCIENCE AND TECHNOLOGY
SCHOOL OF ENGINEERING

Exam	Final Internal Examination 2080		
Level	B. E.	F M	100
Programme	Computer	PM	45
Year/Part	2 nd Year (3 rd Semester)	Time	3 hours

Candidates are required to give their answer in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all questions.

1. a) Evaluate the integral $\int_0^{4a} \int_y^{y^2} \frac{x^2 - y^2}{x^2 + y^2} dx dy$ by changing into polar coordinates. 8
b) Find the volume of the solid cut from the first octant by the surface $z = 4 - x^2 - y$. 7
2. a) Find the solution of the differential equation $y'' + 9y = 0$, by using power series method. 7
b) i) Express $x^3 - 5x^2 + x + 2$ in term of Legendre polynomials. 4
ii) Define Bessel's function of the first kind. Show that:
 $x J'_\nu(x) = \nu J_\nu(x) - x J_{\nu+1}(x)$. 4
3. a) Define Laplace Transform of a function. Using Laplace transform prove the following: 8
i. $\mathcal{L}\{t e^{-4t} \sin 3t\} = \frac{6(s+4)}{[(s+4)^2 + 9]^2}$
ii. $\mathcal{L}^{-1}\left\{\frac{1}{s^2(s^2 + w^2)}\right\} = \frac{1}{w^2} \left(t - \frac{\sin wt}{w}\right)$
OR
State and prove second shifting theorem of Laplace transform. Use it to find the inverse of $\frac{e^{-2s}}{s^2}$. 7
b) Solve by using the Laplace transform.
 $y'' + 4y' + 4y = e^{-t}$, where $y(0) = 0, y'(0) = 0$.

4. a) A particle moves on the curve $x = t^3 + 1, y = t^2, z = 2t + 5$ where t is the time. Find the component of its velocity and acceleration at $t = 1$ in the direction $\vec{i} + \vec{j} + 3\vec{k}$ 7
b) Define Divergence and Curl of a vector. If $\phi = \log(x^2 + y^2 + z^2)$ find $\text{div}(\text{grad } \phi)$ and $\text{curl}(\text{grad } \phi)$. 8
5. a) Verify Green's theorem for $\vec{F} = (x - y)\vec{i} + (x + y)\vec{j}$ over the region R bounded by $y = x^2$ and $x = y^2$. 7
b) State Gauss divergent theorem. Use divergent theorem to evaluate, $\iint_S \vec{F} \cdot \vec{n} dA$, where (e^x, e^y, e^z) , S is the surface of the cube $|x| \leq 1, |y| \leq 1$, and $|z| \leq 1$. 8
OR
State Stoke's theorem. Find $\oint_C \vec{F} \cdot d\vec{r}$ counter clockwise around the boundary C of the region R , where $\vec{F} = (\sin y, \cos x)$, R is the triangle with vertices $(0,0), (\pi, 0)$ and $(\pi, 1)$
6. a) Obtain the Fourier series for the function $f(x) = x^2$ in the interval $0 < x < 2\pi$. 8
b) Define non-linear advection equation. Find the breaking time for $u_t + 2u \cdot u_x = 0, u(x, 0) = e^{-x^2}$. 7
7. Attempt all the questions: $[4 \times 2.5 = 10]$
a) Define unit step function and find its Laplace transform.
b) Evaluate the integral $\int_{(0,0)}^{(1, \frac{\pi}{2})} [e^x (\cos y dx - \sin y dy)]$.
c) Define periodic function. Find period of the function $f(x) = \cos 5x$.
d) Find the Fourier sine series of function $f(x) = 1$ for $0 < x < L$.



*****The End*****

Candidates are required to give answers in their own words as far as practicable.
The figure in the margin indicates full marks.
Attempt all the questions

1. a. Evaluate the given integral: $\int_0^2 \int_{x^2}^4 y \cos(x^2) dx dy$

7

b. Find the volume in the first octant bounded by the co-ordinate planes, the cylinder $x^2 + y^2 = 4$ and the plane $z + y = 3$.

8.

2. a. Solve by using power series: $y'' - 4xy' + (4x^2 - 2)y = 0$.

7.

b. (i) Define the Legendre's Equation of order n. Write the formula of $P_n(x)$ and sketch the graph of $p_2(x)$.

(ii) Show that $J_1(x) = \sqrt{\frac{2}{\pi x}} \sin x$.

4+4

3. a. State and prove the Second shifting theorem of Laplace Transform. Find the Laplace transform of $te^{2t} \sin t$.

4+3

b. Solve the given initial value problem using the Laplace transform

$y'' + 4y' + 3y = e^{-t}$. $y(0) = y'(0) = 1$

8

4. a. If $\vec{V} = \frac{x\vec{i} + y\vec{j} + z\vec{k}}{\sqrt{x^2 + y^2 + z^2}}$ Show that: $\nabla \cdot \vec{V} = \frac{2}{\sqrt{x^2 + y^2 + z^2}}$ and $\nabla \times \vec{V} = 0$.

8

b. Prove that: $\vec{F} = r^2 \vec{r}$, Show that $\vec{F}$ is a conservative vector field and scalar potential is

$\phi = \frac{r^4}{4} + \text{Constant}..$

OR

Evaluate: $\int_{(0,1,\frac{1}{2})}^{(\frac{\pi}{2},3,2)} [y^2 \cos x dx + (2y \sin x + e^{2z}) dy + 2ye^{2z} dz]$

8

5. a. Evaluate by using Green's Theorem of $\oint [(y - \sin x) dx + \cos x dy]$ where C is the triangle

with vertices $(0,0)$, $(\frac{\pi}{2}, 0)$ and $(\frac{\pi}{2}, 1)$.

7

b. Find $\iint_S (\vec{F} \cdot \vec{n}) ds$, for $\vec{F} = x^2 \vec{i} + y^2 \vec{j} + z^2 \vec{k}$, $\vec{r} = (u \cos v, u \sin v, 3v); 0 \leq u \leq 1, 0 \leq v \leq 2\pi$.

OR

Evaluate $\oint_C \vec{F} \cdot d\vec{r}$ by using Stoke's Theorem where $\vec{F} = y \vec{i} + xz \vec{j} - zy^3 \vec{k}$, and

$$C: x^2 + y^2 = 4, z = -3.$$

8

6. a. Find the Fourier cosine series of the function $\vec{F} = x^2$ for $0 < x < \pi$ in half range. Also

show that $1 + \frac{1}{4} + \frac{1}{9} + \frac{1}{16} + \dots = \frac{\pi^2}{6}$.

8

b. Derive one dimensional traffic flow model using conservation law.

7

7. Attempt the following

$$4 \times 2.5 = 10$$

a. Find the inverse of Laplace transforms of $\frac{1}{(s^2 + w^2)s^2}$.

b. Using the divergence theorems to find $\iiint_S (\vec{F} \cdot \vec{n}) ds$, where $\vec{F} = e^x \vec{i} + \vec{j} + e^z \vec{k}$ and

$$S: 0 \leq x \leq 1, 0 \leq y \leq 1, 0 \leq z \leq 1$$

c. Find the Fourier series of the function $f(x) = x$ in the interval $-\pi < x < \pi$

d. Find the a_n of the Fourier Series of $f(x) = \frac{\pi - x}{2}$ for $0 < x < 2$.

The end

Candidates are required to give answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

1. a. Evaluate the given integral: $\int_0^1 \int_0^{1-x} \frac{xydydx}{\sqrt{x^2+y^2}}$ (7) 7

- b. Find the volume bounded by the xy-planes, the cylinder $x^2 + y^2 = 1$ and the plane $x+y+z=3$. (8) 8

2. a. Solve the Bessel's equation: $xy'' + xy' + (x^2 - n^2)y = 0$. By using Frobenius method. 7

- b. (i) Solve by power Series: $y' = 3y$. (4) 4

- (ii) If $P_n(x) = \frac{1}{2^n n!} \frac{d^n [(x^2 - 1)^n]}{dx^n}$ then show that: $P_n(1) = 1$ and $P_n(-1) = (-1)^n$. 4+4

3. a. Define Convolution of two functions. State and prove Convolution Theorem.

OR

- (i) Define Unit step function. Find the Laplace transform of $L\{\sin 2t \cdot u(t - \pi)\}$.

- (ii) Find the inverse Laplace transform of $\frac{s+1}{s^2(s^2+1)}$. (5) 5

- b. Solve the given initial value problem using the Laplace transform

$$y'' + y' - 2y = t, \quad y(0) = 1, \quad y'(0) = 0$$

(2) 8

- a. The necessary and sufficient condition for the vector value function $\vec{a}$ of the scalar variable t

to have a constant direction is $\vec{a} \times \frac{d\vec{a}}{dt} = 0$. (7) 7

- b. If $\phi = \log(x^2 + y^2 + z^2)$ then find $\text{div}(\text{grad } \phi)$ and $\text{curl}(\text{grad } \phi)$. (2) 8

(2)

- a. Calculate $\int_C \vec{F} \cdot d\vec{r}$ where $\vec{F} = (\cosh x, \sinh y, e^z)$ and C be a path given by

$$\vec{r} = (t, t^2, t^3) \text{ From } (0,0,0) \text{ to } (2,4,8).$$

OR Evaluate by using Green's Theorem of $\oint_C x^2 y^2 dx - y^2 x^2 dy$ where C is the boundary of the region with vertices (1,1), (2,2) and (1,1).

b. Evaluate $\oint_C \vec{F} \cdot d\vec{r}$ by using Stoke's Theorem where $\vec{F} = y^3 \vec{i} + x^3 \vec{k}$, and C is the boundary of the triangle with vertices (1,0,0), (0,1,0), (0,0,1).

6. a. Find the Fourier cosine series of the function $f(x) = \begin{cases} x & \text{if } -\frac{\pi}{2} < x < \frac{\pi}{2} \\ \pi - x & \text{if } \frac{\pi}{2} < x < \frac{3\pi}{2} \end{cases}$

b. Define advection equation, non-linear advection equation and Breaking time. Find the breaking time for $u_t + u u_x = 0$, $u(x,0) = e^{-x^2}$

7. Attempt the following

4 × 2.5 = 10

a. Find the inverse of Laplace transforms of $\frac{s^2}{(s^2 + a^2)(s^2 + b^2)}$.

b. Evaluate: $\iiint_V (x - y - z) dx dy dz$, where $V: 1 \leq x \leq 2, 2 \leq y \leq 3, 1 \leq z \leq 3$.

c. Find the Fourier sine series of the function $f(x) = x$ in the interval $0 < x < \pi$

d. Find the a_n of the Fourier Series of $f(x) = \frac{\pi - x}{2}$ for $-\pi < x < \pi$.

The end

4. a) A particle moves on the curve $x = t^3 + 1$, $y = t^2$, $z = 2t + 5$ where t is the time. Find the component of its velocity and acceleration at $t = 1$ in the direction $\vec{i} + \vec{j} + 3\vec{k}$ 7

b) Define Divergence and Curl of a vector. If $\phi = \log(x^2 + y^2 + z^2)$ find $\text{div}(\text{grad } \phi)$ and $\text{curl}(\text{grad } \phi)$. 8

5. a) Verify Green's theorem for $\vec{F} = (x - y)\vec{i} + (x + y)\vec{j}$ over the region R bounded by $y = x^2$ and $x = y^2$. 7

b) State Gauss divergent theorem. Use divergent theorem to evaluate, $\iint_S \vec{F} \cdot \vec{n} dA$, where (e^x, e^y, e^z) , S is the surface of the cube $|x| \leq 1$, $|y| \leq 1$, and $|z| \leq 1$. 8

OR

State Stoke's theorem. Find $\oint_C \vec{F} \cdot d\vec{r}$ counter clockwise around the boundary C of the region R , where $\vec{F} = (\sin y, \cos x)$, R is the triangle with vertices $(0,0)$, $(\pi, 0)$ and $(\pi, 1)$

6. a) Obtain the Fourier series for the function $f(x) = x^2$ in the interval $0 < x < 2\pi$. 8

b) Define non-linear advection equation. Find the breaking time for $u_t + 2u \cdot u_x = 0$, $u(x, 0) = e^{-x^2}$. 7

7. Attempt all the questions: $[4 \times 2.5 = 10]$

a) Define unit step function and find its Laplace transform.

b) Evaluate the integral $\int_{(0,0)}^{(1, \frac{\pi}{2})} [e^x (\cos y dx - \sin y dy)]$.

c) Define periodic function. Find period of the function $f(x) = \cos 5x$.

d) Find the Fourier sine series of function $f(x) = 1$ for $0 < x < L$.



*****The End*****



POKHARA UNIVERSITY
FACULTY OF SCIENCE AND TECHNOLOGY
SCHOOL OF ENGINEERING

Exam	Final Internal Examination 2080		
Level	B. E.	F M	100
Programme	Computer	PM	45
Year/Part	2 nd Year (3 rd Semester)	Time	3 hours

Candidates are required to give their answer in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all questions.

1. a) Evaluate the integral $\int_0^{4a} \int_y^{4a} \frac{x^2 - y^2}{x^2 + y^2} dx dy$ by changing into polar coordinates. 8
7
 b) Find the volume of the solid cut from the first octant by the surface $z = 4 - x^2 - y$.
2. a) Find the solution of the differential equation $y'' + 9y = 0$, by using power series method. 7
 b) i) Express $x^3 - 5x^2 + x + 2$ in term of Legendre polynomials. 4
 ii) Define Bessel's function of the first kind. Show that:

$$xJ'_v(x) = vJ_v(x) - xJ_{v+1}(x).$$
 4
3. a) Define Laplace Transform of a function. Using Laplace transform prove the following: 8

i. $\mathcal{L}\{t e^{-4t} \sin 3t\} = \frac{6(s+4)}{[(s+4)^2 + 9]^2}$

ii. $\mathcal{L}^{-1}\left\{\frac{1}{s^2(s^2 + w^2)}\right\} = \frac{1}{w^2}\left(t - \frac{\sin wt}{w}\right)$

OR

State and prove second shifting theorem of Laplace transform. Use it to find the inverse of $\frac{e^{-2s}}{s^2}$.

- b) Solve by using the Laplace transform.
 $y'' + 4y' + 4y = e^{-t}$, where $y(0) = 0, y'(0) = 0$.

UNITED TECHNICAL COLLEGE
Semester-Fall

Level : Bachelor
Programme : BEE/Comp-III
Course : Calculus-II

Year : 2023
Full Marks : 100
Pass Marks : 45
Time : 1.5hrs

Candidates are required to give their answers in their own words as far as practicable.

Attempt the Questions.

- 1 a Show that the volume of solid enclosed between the cylinder $x^2 + y^2 = 2ax$ and $z^2 = 2ax$ is $\frac{64a^3}{15}$ Cu.unit.
b Evaluate: $\iiint x^2 y z dx dy dz$ over the volume of tetrahedron bounded by $x = 0, y = 0, z = 0$ and $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$.
- 2 a Change the Cartesian Integrals $\int_{-a}^a \int_0^{\sqrt{a^2-x^2}} e^{-(x^2+y^2)} dy dx$ to polar form and Integrate the polar Integrals.
b Applying Power Series Method. Solve the differential equation $y'' - y' = 0$.
- 3 a Define Legendre Equation and find its solution.
b Show that, $xxJ'_n(x) = -nJ_n(x) + xJ_{n-1}(x)$
- 4 a Find the Laplace Transform of
i. $\frac{e^{-at} - e^{-bt}}{t}$
ii. $t^2 \cosh wt$
b Solve the initial value problem by Laplace Transform.
 $y'' + 2y' + 5y = 0, y(0) = 2, y'(0) = -4$.
- 5 a Define Convolution of two functions. Find $L^{-1}\left(\frac{1}{s^2(s-1)}\right)$ by using Convolution Theorem.
b Find the Fourier series of $f(x) = |x|$ for $-\pi < x < \pi$ and show that $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} \dots = \frac{\pi^2}{8}$.
- 6 a Find the Fourier sine and Cosine series of the function $f(x) = x^2$ for $0 < x < L$.

- b Determine the Constants a and b such that the curl of $\vec{F} = (2xy + 3yz)\vec{i} + (x^2 + axz - 4z^2)\vec{j} - (3xy + byz)\vec{k}$ is zero

7 Write Shorts Notes (Any One)

- a Find $\oint_C \vec{f} \cdot d\vec{r}$, where $\vec{F} = \left(y, \frac{z}{2}, \frac{3y}{2}\right)$ C is circle $x^2 + y^2 + z^2 = 6z$ and $z = x + 3$.

- b Find $\iint_S \vec{f} \cdot \vec{n} dA$, where $\vec{F} = (1, xy, yz)$ and S is surface $x^2 + y^2 \leq z, y \geq 0, z \leq 4$.

- 8 a Derive the Conservation Law. Solve $u_t + 2u \cdot u_x = 0$ where $u(x, 0) = e^{-x}$.

- b Derive minimum time $t_b = -\frac{1}{f'(\lambda)}$. Find the breaking time for $u_t + 2u \cdot u_x = 0, u(x, 0) = e^{-x^2}$

NEPAL COLLEGE OF INFORMATION TECHNOLOGY
Assessment Fall 2023

Level: Bachelor

Year : 2024

Programme: BE-IT_M&D_III

Full Marks : 100

Course: Calculus-II

Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

- 1 (a) Evaluate $\int_0^{2a} \int_0^{\sqrt{2ax-x^2}} (x^2 + y^2) dy dx$ By changing into polar coordinates. 5
- (b) Evaluate $\int_0^{\log 2} \int_0^x \int_0^{x+\log y} e^{x+y+z} dz dy dx$ 5
- (c) State Dirichlet's integral, use it to find the volume of the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$. 5
- 2 (a) Solve the following differential equation $y'' + 4y = 0$ by power series method. 5
- (b) If $P_n(x)$ and $P_m(x)$ are Legendre polynomials, then prove that $\int_{-1}^1 P_n(x) P_m(x) dx = 0$, for m not equal to n . 5

OR

Express $5x^2 + 5x - 3$ into Legendre polynomials.
- (c) Define Bessel's function of first kind of order n $J_n(x)$ and $J_{-n}(x)$. Show that $[J_{1/2}(x)]^2 + [J_{-1/2}(x)]^2 = \frac{2}{\pi x}$. 5

OR

Show that $x J_n'(x) = n J_n(x) - x J_{n+1}(x)$
- 3 (a) (i) Show that $\int_0^{\infty} \frac{e^{-2t} \sin 2t}{t} dt = \frac{\pi}{2}$ By using Laplace Transform. 4

(II) Define convolution of two function. Find the inverse Laplace transform of $\frac{1}{s(s^2+4)}$ 4

(b) Solve the following initial value problem using Laplace Transform: 7

$$y'' - 2y' + y = \quad, y(0) = 2, y'(0) = -1$$

4 (a) A moving particle along the curve $\vec{r} = (t^3 - 4t)\vec{i} + (t^2 + 4t)\vec{j} + (8t^2 - 3t^3)\vec{k}$, where t is time, find the magnitude of tangential component of its acceleration at $t=2$. 5

(b) Define the gradient of a scalar function with geometrical interpretation. Find the directional derivative of $x^2y^2z^2$ at the point $(1, 1, -1)$ in the direction of the tangent $e^t, y = \sin 2t + 1, z = 1 - \cos t$ at $t = 0$. 5

(c) Define line integral. Determine whether the line integral $\int (2xyz^2)dx + (x^2z^2 + z \cos yz)dy + (2x^2yz + y \cos yz)dz$ is independent of the path of integration? If so, then evaluate it from $(1, 0, 1)$ to $(0, \frac{\pi}{2}, 1)$. 5

OR

Show that the value under the integral sign

$$\int_{(0, \frac{\pi}{2}, 1)}^{(0, \frac{\pi}{2}, 1)} (-z \sin xz dx + \cos y dy -$$

$x \sin xz) dz$ is exact and evaluate the integral.

5 (a) Green's theorem for plane. Use the Green's theorem in the plane for $\oint_C (3x^2y - 8y^2)dx + (4x^2 - 6xy)dy$, where C is the boundary of the region defined by $y^2 = x^2$. 5

(b) Define Stoke's theorem. Evaluate $\oint_C \vec{F} \cdot d\vec{r}$ by Stoke's theorem where $\vec{F} = (x^2 + y - 4)\vec{i} + 3xy\vec{j} + (2xz + z^2)\vec{k}$ over the surface of hemisphere $x^2 + y^2 + z^2 = 16$ above the xy -plane. 5

OR

Evaluate $\iint_S \vec{F} \cdot \vec{n} ds$ where $\vec{F} = (x+y^2)\vec{i} - 2x\vec{j} + 2yz\vec{k}$ and S is the surface of the plane $2x+y+2z=6$ in the first octant.

c) Define Gauss Divergence theorem. Evaluate $\iint \vec{F} \cdot \hat{n} \, ds$ by using Gauss divergence theorem where $\vec{F} = (e^x, e^y, e^z)$, $S: -1 \leq x, y, z \leq +1$. 5

6 (a) Find the Fourier cosine series of the function $f(x) = L-x$ for $0 < x < L$. 7

(b) Show that the fourier series A periodic function $f(x)$ of period 2π is defined for $-\pi < x < +\pi$ as follows: 8

$$f(x) = 2 \cos x, \text{ for } |x| < \frac{\pi}{2}$$

0, otherwise

of $f(x)$ is given by

$$f(x) = \frac{4}{\pi} \left[\frac{1}{2} + \frac{\pi}{4} \cos x + \frac{1}{1.3} \cos 2x - \frac{1}{3.5} \cos 4x + \frac{1}{5.7} \cos 6x - \dots \right]$$

7 (a) Solve: 5

$$y^2 u_x - xy u_y = x(u - 2y),$$

where x and y are independent variable and u is dependent

Define non-linear Advection equation. Find the breaking time for 5

(b) $u_t + 2u u_x = 0, u(x, 0) = e^{-x^2}$

National Academy of Science and Technology

(Affiliated to Pokhara University)

Dhangadhi, Kailali

Pre-University Examination

Level: Bachelor

Semester : III_Fall

Year: 2023

Program: B.E. (Computer/Civil)

F.M.: 100

Course: Calculus II

P. M.: 45

Time : 3 Hrs.

Candidates are required to give their answer in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Evaluate $\int_0^\pi \int_x^\pi \frac{\sin y}{y} dy dx$. [7]

b) State Dirichlet's theorem. Use it to evaluate the triple integral

$\iiint_V x^2 dx dy dz$. $\checkmark$. $x \geq 0, y \geq 0, z \geq 0 \Rightarrow \frac{x}{a} + \frac{y}{a} + \frac{z}{a} = 1$ [8]

2. a) Using series solution method, solve: $(1-x^2)y'' - 2xy' + 2y = 0$. [8]

b) Express $x^3 + 2x^2 - x - 3$ in terms of Legendre polynomials. [7]

OR

Show that $\frac{d}{dx} [x^{-n} J_n(x)] = -x^{-n} J_{n+1}(x)$.

3. a) Evaluate: [8]

i) $L \left\{ \frac{e^{-t} \sin at}{t} \right\}$ ii) $L^{-1} \left\{ \frac{1}{(s-1)(s^2+9)} \right\}$

b) Using Laplace transform, solve the initial value

problem: $y'' - 2y' + y = e^t$, $y(0) = 2$, $y'(0) = -1$. [7]

4. a) Show that the necessary and sufficient condition for the vector valued function $\vec{a}$ of the scalar variable t to have a constant direction

is $\vec{a} \times \frac{d\vec{a}}{dt} = \vec{0}$. [7]

b) A particle moves along the curve $x=t^3-4t$, $y=t^2+4t$, $z=8t^2-3t^3$, where t is the time. Find the component of its velocity and acceleration at time $t=1$ in the direction of $\hat{i} - 3\hat{j} + 2\hat{k}$. [8]

5. a) State Green's theorem in the plane. Use it to evaluate the integral

$\oint_C [(y - \sin x)dx + \cos x dy]$. $\checkmark$ C is the boundary of the R with vertices $(0,0), (\pi,0), (\pi,1)$ [7]

- b) Evaluate the surface integral $\iint_S (\vec{F} \cdot \hat{n}) ds$, where
 $\vec{F} = (x^2, -e^y, 1)$, $S: x + y + z = 1; x \geq 0, y \geq 0, z \geq 0$. [8]

OR

Using Gauss Divergence theorem, Evaluate the surface integral
 $\iint_S (\vec{F} \cdot \hat{n}) ds$, where $\vec{F} = (2xy + z, y^2, -x - 3)$, $S: 2x + 2y + z = 6$;
 $x = 0, y = 0, z = 0$.

6. a) Find the Fourier series of the function $f(x) = x^2, -\pi \leq x \leq \pi$. [8]

Hence show that $\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \dots = \frac{\pi^2}{6}$.

- b) Write the general form of one dimensional conservation law. Write any two applications of conservation law. Solve the p.d.e, $u_t + 2u \cdot u_x = 0$,
 $u(x, 0) = e^{-x}$. [1+1+5=7]

7. Attempt Any four [2.5×4=10]

- Find the Laplace transform of $e^{-3t}u(t-2)$.
- Find the unit normal vector to the surface $f = x^2y + 2xz - 4$ at $(2, -2, 3)$.
- Show that the value under the integral sign of $\int_{(0, \pi)}^{(3, \frac{\pi}{2})} e^x (\cos y dx - \sin y dy)$ is exact. Hence evaluate it.
- Check whether the function is odd or even. Find it's period.

$$f(x) = \begin{cases} \frac{1}{2} + x & \text{for } -\frac{1}{2} < x < 0 \\ \frac{1}{2} - x & \text{for } 0 < x < \frac{1}{2} \end{cases}$$
- Evaluate $\int_0^\pi \int_0^{\sin x} y dy dx$.

MODEL QUESTION

Candidates are required to give their answer in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all questions.

1. a) Evaluate the integral: $\int_0^2 \int_{y^2}^4 y \cos(x^2) dx dy$. 5
- b) Evaluate the integral: $\int_0^1 \int_0^{(1-x)} \int_0^{(x+y)} e^z dz dy dx$. 5
- c) Find the volume in the first octant bounded by the co-ordinate planes, the cylinder $x^2 + y^2 = 1$ and the plane $z + y = 3$. 5
2. a) Solve by using power series: $y'' - 4xy' + (4x^2 - 2)y = 0$. 7
- b) (i) Express: $2x^2 - 4x + 2$ as Legendre polynomial. 4+4
- (ii) Show that: $J_{\frac{1}{2}}(x) = \sqrt{\frac{2}{\pi x}} \sin x$.
- OR
- Find the solution of Bessel's Equation.
- $$x^2 y'' + xy' + (x^2 - n^2)y = 0$$
3. a) (i) State first shifting theorem of Laplace Transform and find the Laplace transform of $t \cos at$. 4+4
- (ii) Find the inverse Laplace transform of the function $\frac{s+1}{s^2(s+3)}$.
- b) Apply Laplace transform to solve the initial value problem 7
- $$y'' + 4y' + 3y = e^{-t}, \quad y(0) = y'(0) = 1$$

4. a) A particle moves along the curve $(t^3 + 1, t^2, 2t + 5)$. Find the component of the velocity and acceleration at $t=1$ along $\vec{i} + \vec{j} + 3\vec{k}$. 5
- b) If $\phi = \ln(x^2 + y^2 + z^2)$, then find $\text{grad}\phi$ and $\text{div}(\text{grad}\phi)$. 5
- c) Evaluate $\oint_C (x^2 - 3y)dx + (x + \sin y)dy$ where C is the boundary of the triangle with vertices $(0,0)$, $(1,0)$ and $(0,2)$. 5
5. a) Find $\iint_S (\vec{F} \cdot \vec{n})ds$, for $\vec{F} = x^2 \vec{i} + y^2 \vec{j} + z^2 \vec{k}$, 7
- $\vec{r} = (u \cos v, u \sin v, 3v); 0 \leq u \leq 1, 0 \leq v \leq 2\pi$.
- b) Evaluate $\oint_C (\vec{F} \cdot d\vec{r})$ by using Stoke's theorem, where $\vec{F} = (y, xz^3, -zy^3)$ and $C: x^2 + y^2 = 4, z = 3$. 8
- OR**
- State Gauss divergence theorem and use it to evaluate the surface integral $\iint_S \vec{F} \cdot \hat{n} ds$ for $\vec{F} = 4xz\hat{i} - y^2\hat{j} + yz\hat{k}$ and S is a cube $0 \leq x \leq 1, 0 \leq y \leq 1$ and $0 \leq z \leq 1$.
6. a) Find the Fourier series of $f(x) = \frac{x^2}{2}$ for $-\pi \leq x \leq \pi$ and deduce that 7
- $1 + \frac{1}{4} + \frac{1}{9} + \frac{1}{16} + \dots = \frac{\pi^2}{6}$.
- b) Find Fourier half range cosine and sine series of $f(x) = e^x$ in $(0, L)$. 8
7. Attempt any two questions: **[2 × 5 = 10]**
- a) Derive one dimensional traffic flow model using conservation law.
- b) Find the breaking time for $u_t + u u_x = 0$, $u(x,0) = e^{-x^2}$.
- c) Evaluate $\int_{(1,0,2)}^{(-2,1,3)} [(6xy^3 + 2z^2)dx + 9x^2y^2dy + (4xz + 1)dz]$.



Term Test II

Date:	2080/10/14	Full Marks	
Level	BE	Time	1.5 hrs
Programme	BEIT, BCE, BCV		
Semester	III		

Subject: - Calculus II

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
- ✓ The figures in the margin indicate Full Marks.
- ✓ Assume suitable data if necessary.

1. State Stoke's Theorem. Using stokes theorem evaluate $\int_C \vec{F} \cdot d\vec{r}$ where $\vec{F} = (y, \frac{z}{2}, \frac{3y}{2})$ and C is the ellipse $x^2 + y^2 + z^2 = 6z, z = x + 3$. [7]
2. State Gauss Divergence Theorem. Using the divergence theorem find $\iiint_S \vec{F} \cdot \hat{n} ds$ where $\vec{F} = y^2 e^z \hat{i} - xy \hat{j} + x \tan^{-1} y \hat{k}$ and S is the portion of the plane $2x + 3y + z = 6$ in the first octant. [8]
3. Evaluate the integral $\int_0^2 \int_{-2}^2 y \cos x^2 dx dy$ by changing the equivalent integral obtained by reversing the order of integration. [7]
4. Find the volume bounded by the xy -plane, the paraboloid $2z = x^2 + y^2$ and the cylinder $x^2 + y^2 = 4$. [8]

OR

Find the Fourier series of $f(x) = x - x^2$ if $-\pi < x < \pi$ and show that

$$\frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \dots = \frac{\pi^2}{12}$$

5. Solve the equation $y' = y$ by power series method. [7]
6. State Green's Theorem. Using it evaluate $\oint_C [(x^3 - 3y)dx + (x + \sin y)dy]$, C : the boundary of the triangle with vertices $(0,0), (1,0), (0,2)$. [8]
7. Attempt any two. [2 x 2.5 = 5]
 - a) Define Periodic function with period. Find the period of $\tan\left(\frac{3x}{5}\right)$.
 - b) Define Dirichlet integral and state Dirichlet Theorem.
 - c) Find the area of the circle $x^2 + y^2 = a^2$ by changing to polar co-ordinates using double integral.

$$(x_1, y_1) = (1, 0)$$

$$(x_2, y_2) = (0, 2)$$

$$y - y_1 = m(x - x_1)$$

$$y = -2(x - 1)$$

$$y = 2 - 2x$$



Pokhara University
Everest Engineering College
Final Internal Assessment
Fall-2023

Level: Bachelor

F.M: 100

Program: BE CMP

P.M: 45

Faculty: Science and Technology

Time: 3 hrs.

Subject: Calculus-II (3rd Semester)

Attempt all the questions.

1. (a) Find the volume in the first octant bounded by co-ordinates planes, the cylinder $x^2 + y^2 = 4$ and plane $z + y = 3$. [7]
(OR)
Find by triple integration, volume of sphere $x^2 + y^2 + z^2 = a^2$. [7]
- (b) Evaluate the integral $\int_0^a \int_{y^2/4a}^y \frac{x^2 - y^2}{x^2 + y^2} dx dy$ by changing into polar co-ordinates. [8]
2. (a) Solve by power series method: $(1 - x)y' = y$. [7]
(b) Define Legendre's equation. Also derive the solution of Legendre's equation. [1+7]
3. (a) Evaluate the following:
(i) $\mathcal{L}(t^2 \cos \omega t)$ [4]
(ii) $\mathcal{L}^{-1}\left(\frac{1}{s^2(s^2 + \omega^2)}\right)$ by using convolution. [4]
(b) Solve the differential equation $y'' - 2y' + y = e^t$ $y(0) = 2, y'(0) = -1$ by using Laplace transform. [7]
4. (a) Find the directional derivative of $f = xy^2 + yz^3$ at $(2, -1, 1)$ along the direction of the normal to the surface $x \log z - y^2 + 4 = 0$ at $(-1, 2, 1)$. [7]
(b) If $\vec{v} = x^2yz\vec{i} + xy^2z\vec{j} + xyz^2\vec{k}$ then find [4+4]
(i) $\text{curl}(\text{curl } \vec{v})$
(ii) $\text{div}(\text{curl } \vec{v})$
5. (a) State Stoke's theorem. Evaluate $\oint_C \vec{F} \cdot d\vec{r}$, where $\vec{F} = (z, x, y)$ and S is the hemisphere $z = (a^2 - x^2 - y^2)^{1/2}$. [1+7]
(b) Evaluate $\iint_S \vec{F} \cdot \vec{n} ds$ where $\vec{F} = (x^2, -e^y, 1)$ and S is the surface of plane $x + y + z = 1$ in first octant. [7]
(OR)
Evaluate $\iint_S \vec{F} \cdot \vec{n} ds$ where $\vec{F} = e^x\vec{i} + e^y\vec{j} + e^z\vec{k}$ and $S : |x| \leq 1, |y| \leq 1, |z| \leq 1$ by using Gauss divergence theorem. [7]

6. (a) Define periodic function with example. Find the fourier series representation of the function $f(x) = \frac{x^2}{2}$ for $-\pi < x < \pi$ and then show that: [1+6+1]

$$1 + \frac{1}{4} + \frac{1}{9} + \frac{1}{16} + \dots = \frac{\pi^2}{6}.$$

- (b) Derive the partial differential equation governing the conversation law. [7]

7. Attempt all the questions: [4 × 2.5 = 10]

- (a) Find the Laplace transform of $t^2 u(t-2)$, where u is unit function.
- (b) Show that the value under the integral sign $\int_{(0,1)}^{2,3} [(2x+y^3)dx + (3xy^2+4)dy]$ is exact and evaluate the integral.
- (c) Obtain the half-range fourier sine series of the function $f(x) = \pi - x$ for $0 < x < \pi$.
- (d) Find the general solution of $u_x + u_y = u$.

The End